

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

from skimage.feature import hog
from skimage import exposure

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
image = cv2.imread("Babe.jpg")

if image is None:
    print("Image not found. Check the file path.")
else:
    print("Image loaded successfully!")

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6,6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Image loaded successfully!

Original Image



```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Histogram Equalization
gray_eq = cv2.equalizeHist(gray)

plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(gray_eq, cmap='gray')
plt.title("Preprocessed Image")
plt.axis("off")

plt.show()
```

Grayscale Image



Preprocessed Image



```
sift = cv2.SIFT_create()

keypoints, descriptors = sift.detectAndCompute(gray_eq, None)

print("Number of Keypoints:", len(keypoints))
print("Descriptor Shape:", descriptors.shape)

sift_image = cv2.drawKeypoints(
    image_rgb,
    keypoints,
    None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)

plt.figure(figsize=(10,8))
plt.imshow(sift_image)
plt.title("SIFT Keypoints")
plt.axis("off")
plt.show()
```

Number of Keypoints: 1957
Descriptor Shape: (1957, 128)

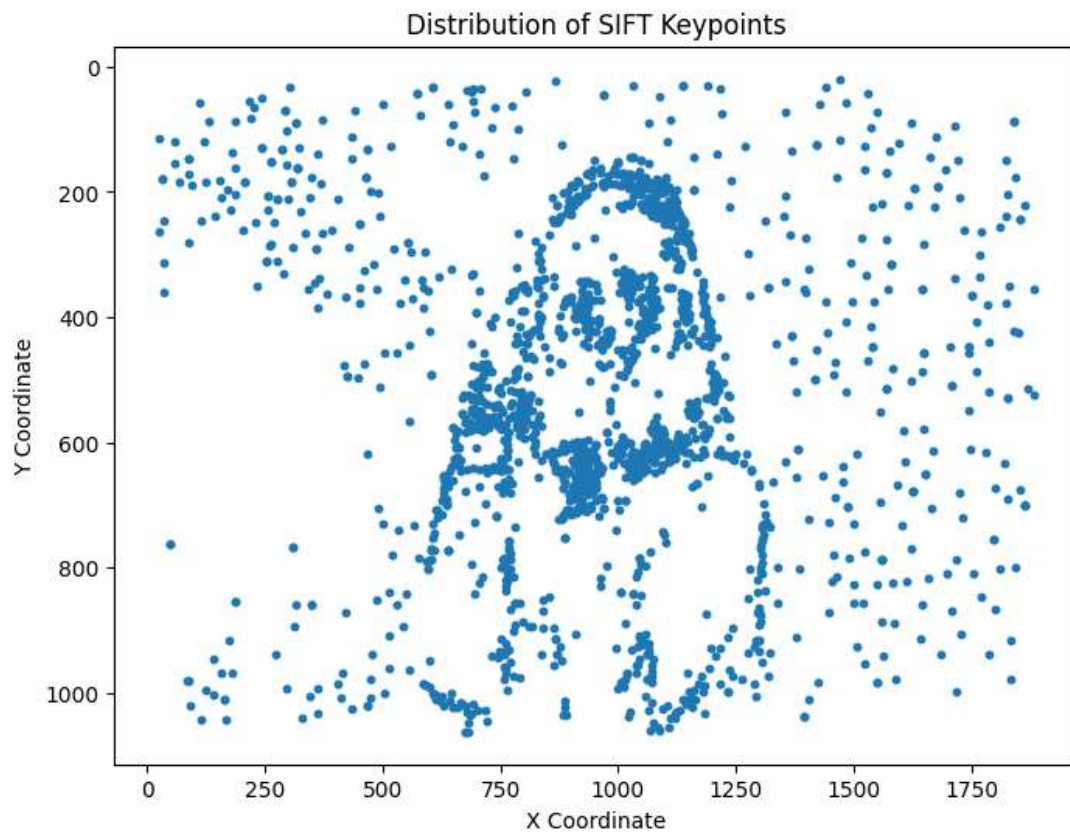
SIFT Keypoints



```
x_coords = [kp.pt[0] for kp in keypoints]
y_coords = [kp.pt[1] for kp in keypoints]

plt.figure(figsize=(8,6))
plt.scatter(x_coords, y_coords, s=10)
plt.gca().invert_yaxis()
plt.title("Distribution of SIFT Keypoints")
plt.xlabel("X Coordinate")
```

```
plt.ylabel("Y Coordinate")
plt.show()
```



```
hog_features, hog_image = hog(
    gray_eq,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    visualize=True,
    block_norm='L2-Hys'
)

print("HOG Feature Vector Length:", len(hog_features))

hog_image_rescaled = exposure.rescale_intensity(
    hog_image,
    in_range=(0, 10)
)

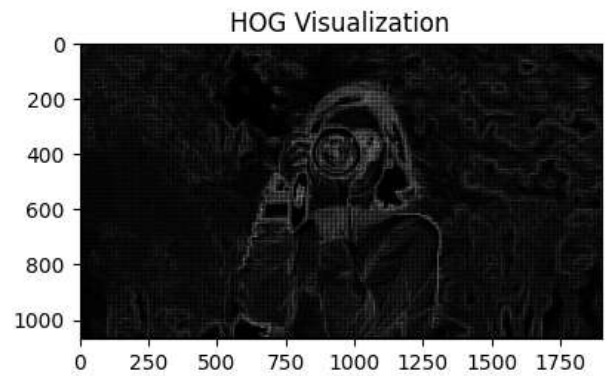
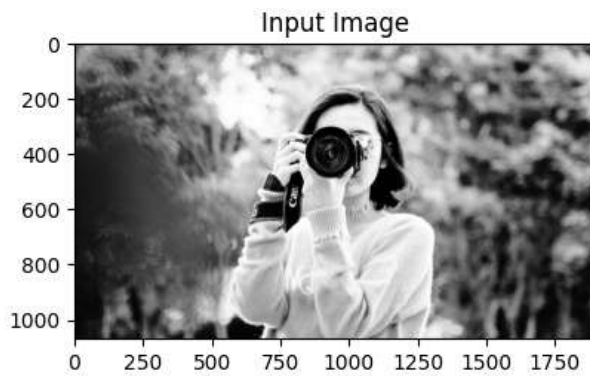
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray_eq, cmap='gray')
plt.title("Input Image")

plt.subplot(1,2,2)
plt.imshow(hog_image_rescaled, cmap='gray')
plt.title("HOG Visualization")

plt.show()
```

HOG Feature Vector Length: 1121472



```
comparison = {
    "Feature Type": ["SIFT", "HOG"],
    "Scale Invariant": ["Yes", "No"],
    "Rotation Invariant": ["Yes", "Limited"],
    "Computation": ["High", "Low"],
    "Application": ["Matching, Recognition", "Detection, Classification"]
}
```

```
for key, value in comparison.items():
    print(f"{key}: {value}")
```

```
Feature Type: ['SIFT', 'HOG']
Scale Invariant: ['Yes', 'No']
Rotation Invariant: ['Yes', 'Limited']
Computation: ['High', 'Low']
Application: ['Matching, Recognition', 'Detection, Classification']
```

```
image2 = cv2.imread("Image2.jpg")

image2_rgb = cv2.cvtColor(image2, cv2.COLOR_BGR2RGB)
gray2 = cv2.cvtColor(image2, cv2.COLOR_BGR2GRAY)

kp1, des1 = sift.detectAndCompute(gray_eq, None)
kp2, des2 = sift.detectAndCompute(gray2, None)

bf = cv2.BFMatcher()

matches = bf.knnMatch(des1, des2, k=2)

good_matches = []

for m, n in matches:
    if m.distance < 0.75 * n.distance:
        good_matches.append(m)

print("Good Matches:", len(good_matches))

matched_image = cv2.drawMatches(
    image_rgb,
    kp1,
    image2_rgb,
    kp2,
    good_matches[:50],
    None,
    flags=2
)

plt.figure(figsize=(15,8))
plt.imshow(matched_image)
plt.title("SIFT Image Matching")
plt.axis("off")
plt.show()
```

Good Matches: 27

SIFT Image Matching

